Bubbles

Your AI conversation co-pilot — live coaching while you talk,
and a smart assistant that remembers.

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THE TEAM & THE BRIEF

Two people. One split-stack build.

Muhammad Ahmad

FA22-BCS-025 · App · Architecture · Frontend

- Flutter client across 6 platforms
- System architecture & data model
- Auth, UI/UX, caching, releases

Attique Rehman

FA22-BCS-164 · AI · Backend · Memory systems

- FastAPI brain & LLM router
- Knowledge graph + vector memory
- Real-time pipeline & workers

IN PLAIN ENGLISH

A friend in your ear — and a memory that lasts.

1

Listens

Hears the conversation as it happens, in real time.

2

Suggests

Quietly whispers what to say next, so you never freeze.

3

Remembers

Keeps your people and topics, so you can ask about them later.

Writing got autocorrect.
Speaking never did.

Speech happens in real time. We notice the wrong word or wrong tone only after the moment is gone.

WHERE THE IDEA CAME FROM

We kept losing the moment.

01

The blank-out

Froze mid-interview — knew the answer, couldn't get it out.

02

The forgotten promise

Couldn't recall what a client actually agreed to last week.

03

The walk-home regret

Realized the better phrasing only hours after it mattered.

WHAT ALREADY EXISTS

Two camps today — neither helps in the moment.

Meeting notetakers

Otter · Fathom · Notta · Gong

Transcribe & summarize — but only after the meeting ends.

General assistants

Siri · Google Assistant · ChatGPT

Answer commands — but forget you between sessions.

THE RESEARCH GAP

Three things nobody does together.

No real-time coaching

Everything on the market is post-meeting analysis.

No long-term context

Assistants forget who and what you talked about.

Not privacy-first

Enterprise tools mine your data for the org, not you.

WHY POST-MEETING ISN'T ENOUGH

The value of advice decays in seconds.

Mistake made

Conversation ends

You read the summary

Feedback only ever lands here → too late to change the outcome.

PROBLEM STATEMENT

Success often depends on communication, yet errors are hard to catch while speaking. Existing AI only transcribes or summarizes after the fact — with no live, context-aware, privacy-respecting feedback.

WHY IT MATTERS

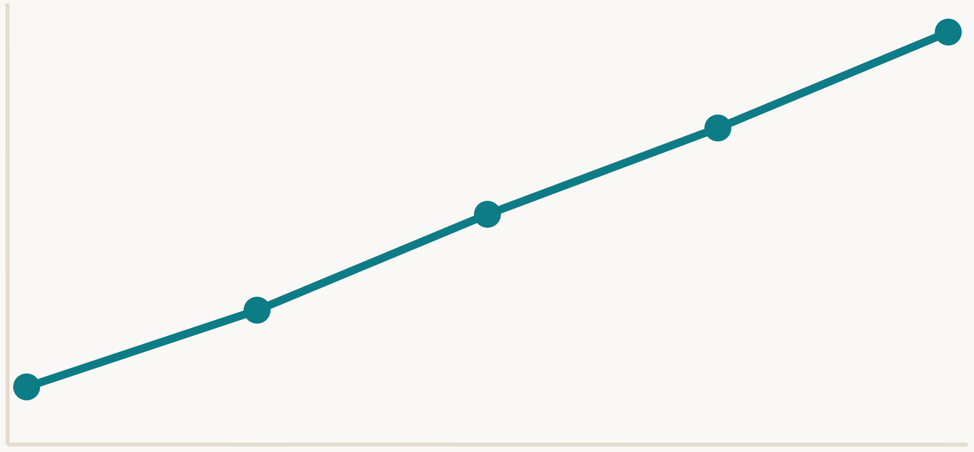
Soft-skills coaching, made a daily habit.

Like Duolingo — but for talking. Track progress, drill weak spots, reward improvement.

SDG 4

Quality Education

Aligned to the UN Sustainable Development Goal we mapped this project to.



Confidence over time

THE SOLUTION, IN FOUR VERBS

Listen → Whisper → Remember → Improve

1
Listen
to the live conversation

›

2
Whisper
the next thing to say

›

3
Remember
the people & topics

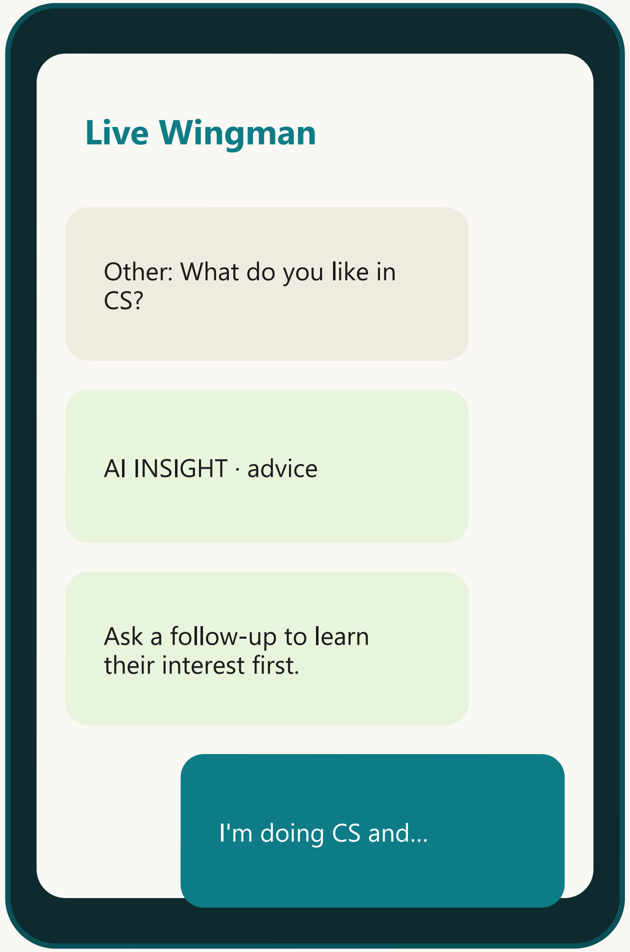
›

4
Improve
from your real mistakes

MODE 1

Live Wingman — coaches you while you talk.

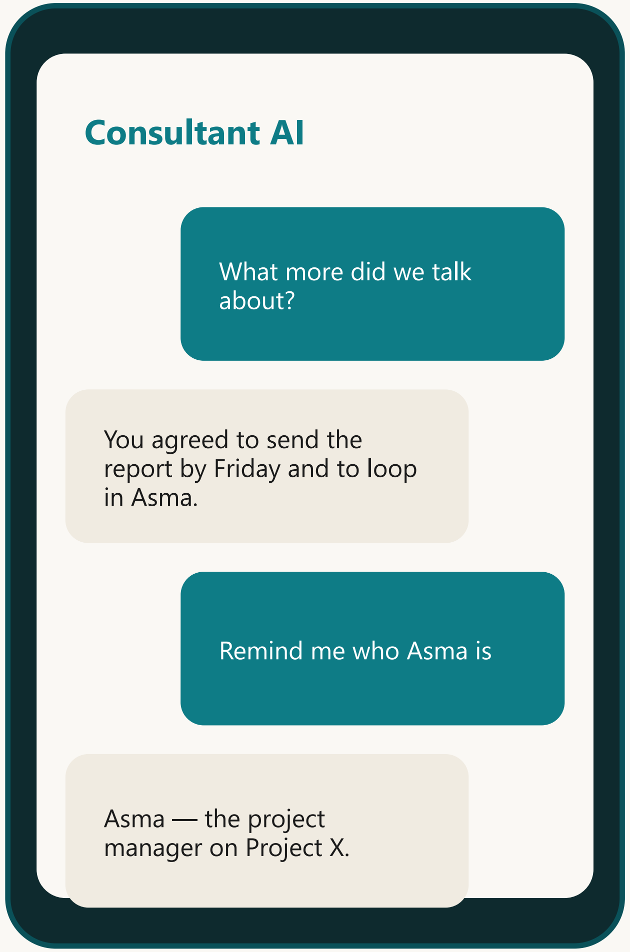
- Real-time advice generated from the live transcript
- Clarifying questions when the moment needs one
- Reacts to the other person — never echoes you back
- Pushed to screen over WebSocket, no flow break



MODE 2

Consultant — ask anything about your past talks.

- Deep, context-aware Q&A over your history
- “What did I promise Client X last week?”
- Streaming, blocking, or batched answers
- Backed by the heavy reasoning model



MODE 3

Speak-Improve — the coaching loop.

1 Mistakes captured ›

2 Turned into drills ›

3 Progress tracked ›

4 Improvement rewarded

↺ *and the loop repeats — every session feeds the next.*

THE DIFFERENTIATOR

Two memories, not one.

Vector memory

Finds semantically similar past moments — fuzzy recall.

"sounds related"

Knowledge graph

Stores factual relationships between people & topics.

User → knows → Asma

Fused before every answer

WHY GRAPHRAG, NOT PLAIN RAG

Plain RAG sounds right. GraphRAG is right.

✕ Plain vector RAG

Retrieves text that reads similar — can confuse two similar events and invent facts.

✓ Graph-augmented RAG

Looks up hard relationships first (ego-graph around your entities) — anchors the answer.

A DELIBERATE TRADE-OFF

Thin client. Heavy brain.

📱 The phone

- Microphone
- Screen
- Battery-light

→

The cloud brain

- All AI: STT, LLM router, memory, graph
- Models stay swappable, not baked into the app
- Same experience on a cheap phone

PART 04

How it actually works.

Three tiers, one clean data path.

ARCHITECTURE AT A GLANCE

Three tiers, top to bottom.

Client — Flutter

Screens · State · Repositories · Cache · Services

Backend — Brain API (FastAPI)

Auth · LLM Router · Endpoints · ARQ Workers

Data & AI

Postgres + pgvector · Redis · Gemini/Cerebras/Groq · LiveKit

THE CLIENT

One codebase → six platforms.

Flutter · Dart ·

Android	**iOS**	**Web**
Windows	**macOS**	**Linux**

25+ screens · 11 repositories · 25 services · offline-first stale-while-revalidate cache over SQLite

Why not native or React Native?

Approach	Verdict	Why
Flutter	✓ Chosen	One Dart codebase, true desktop, 60fps custom UI
Native (×2)	✕	Two-person team can't build & maintain 6 native apps
React Native	✕	JS-bridge overhead, weak desktop story

THE BACKEND BRAIN

Async, top to bottom — 40+ endpoints under /v1.

Sessions	Wingman	Consultant	Voice
Speaker ID	Grammar	Drills	Scenarios
Gamification	Dashboard	Entities	Memory
Persona			

Python 3.12 · FastAPI on Uvicorn/Gunicorn · Pydantic v2 validates every request & response

WHY FASTAPI ASYNC

Real-time audio needs non-blocking I/O.

✕ Sync (Flask/Django)

One worker blocks per audio stream → doesn't scale.

✓ Async (FastAPI)

Await network/LLM calls without freezing the server. Typed schemas + auto OpenAPI for free.

THE LLM ROUTER

Each task → the model that fits it.

Task	Model	Provider chain
Consultant Q&A	gemini-2.5-flash	Gemini → Cerebras → Groq
Wingman (live advice)	llama-3.1-8b	Cerebras → Groq → Gemini
Speech-to-text	whisper-large-v3-turbo	Groq
Embeddings	text-embedding-004	Gemini (ONNX fallback)

Per-provider circuit breakers: a failing provider trips its breaker and we fail over instantly.

WHY MULTI-PROVIDER

No single point of failure.

× One provider

One outage kills the app. One price hike traps you. One model is wrong for some tasks.

✓ Three providers, routed

Right model per job, automatic failover, no vendor lock-in.

VOICE PIPELINE

Wake-word → speech → advice → voice.

1

Wake-word

Porcupine, on-device

›

2

Transport

LiveKit low-latency audio

›

3

STT

Groq Whisper turbo

›

4

Speaker ID

you vs. them

›

5

TTS

Edge-TTS speaks back

Transcript → entities → relationships → graph.

1	2	3	4	5
Transcript	**Extract**	**Model**	**Persist**	**Explore**
live text	fast LLM finds people/topics	NetworkX nodes + edges	Postgres link tables	interactive timeline

BACKGROUND WORKERS (ARQ)

Heavy work happens off the critical path.

HOT PATH · stays fast

Live advice generation — the only thing on the main thread.

BACKGROUND · 12 idempotent jobs

Embeddings · grammar scan · sentiment · knowledge extraction · rolling summaries · session analytics · drill cards · scenarios · achievements · reminders

THE REAL-TIME CHALLENGE

Chained, it took 3–5 seconds.

STT → Vector search → LLM → Response

Conversation needs **< 1 second.** *3–5 s of dead air is useless mid-talk.*

HOW WE BEAT IT

Parallel pipeline + Groq LPU.

Before

~2 s

serial chain, big model

→

After

~300 ms

hot path only + 8B on Groq LPU

Only advice stays on the main thread; knowledge extraction & memory-saving become fire-and-forget.

CHALLENGE: WHO SAID WHAT

Advise on their words — not yours.

Other Can you explain your hobbies?

You I'm doing CS and... stuff.

Other What do you like in CS?

We parse word-level timestamps to split User vs. Other — Wingman reacts to the other person only.

TWO MORE HARD PROBLEMS

Hallucination & lost state.

Hallucination

Force graph facts first, plus a strict prompt: “If it's not in the context, say you don't know.”

Lost state

Stateless HTTP forgets mid-talk. Hold live context in RAM (fast) and persist to Supabase (reliable).

BUILT FOR RELIABILITY

Production guardrails, not a demo script.

- ✓ **Circuit breakers** — isolate a failing provider
- ✓ **Token-bucket rate limiting** — Redis Lua, per user
- ✓ **Retries with back-off** — Tenacity on transient faults
- ✓ **Idempotent jobs** — a retry never double-fires
- ✓ **ULID IDs** — sortable, collision-safe

WHAT'S BUILT

Ten features — all live in the app.

Live Wingman	Consultant	Confidence meter	Spaced-repetition drills	Progress dashboard
Roleplay scenarios	Gamification	Knowledge-graph explorer	Grammar & sentiment	Voice & speaker ID

We'll deep-dive four that show real engineering — then a live demo.

LIVE CONFIDENCE METER

On-device, zero added latency.

Counts filler words (“um”, “uh”, “like”) and hedges (“sort of”, “I guess”) over a rolling 8 seconds — smoothed so it doesn't jitter.

Low	Building	Confident
0.0 – 0.4	0.4 – 0.7	0.7 – 1.0

Server only stores the per-turn score at session end — for the confidence-trend chart.

SPACED-REPETITION DRILLS

Your mistakes → flashcards → 5 Leitner boxes.

Box 1	Box 2	Box 3	Box 4	Box 5
review every	review every	review every	review every	review every
1 d	**3 d**	**7 d**	**14 d**	**30 d**

Correct → up a box (seen less often).   Wrong → back to Box 1.   XP on every transition (+15 / +5), idempotent.

PROGRESS DASHBOARD

"Am I actually improving?"

↓ 34%	↑ 35%	+33%	+23%
Mistakes	**Sentiment**	**Sessions**	**XP**
vs previous window	vs previous window	vs previous window	vs previous window

One endpoint, time-bucketed daily / weekly / monthly — pure read-side aggregation, no new tables.

Practice the conversation before you have it.

Salary review with your manager

Person: Asma (Project Manager)

Difficulty: Hard

Opening: *"Thanks for making time — I'd like to talk about my role."*

Generated from your graph

Pass it → auto-graded against its success criteria → +40 XP. The feed never repeats — used topics are excluded.

GAMIFICATION

Habit mechanics — for talking.

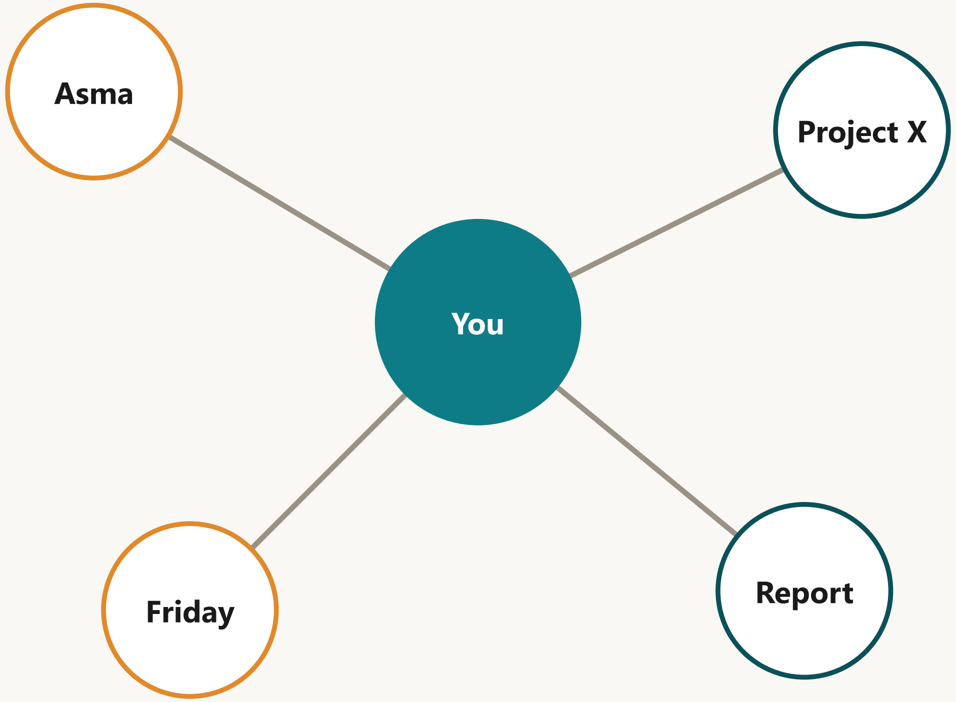
- **Quests** — small daily goals
- **Achievements** — milestone badges
- **Rewards** — redeemable points
- **Streaks** — keep the chain alive
- **Leaderboard** — opt-in only — consistent with privacy-first

KNOWLEDGE-GRAPH EXPLORER

Your memory, made tangible.

Browse the people, topics and entities you've mentioned — on a timeline.

The same nodes the AI uses for grounding — now yours to explore.



DATA MODEL

Three pillars in Postgres.

Vector memory

384-dim embeddings for semantic recall (RAG).

Knowledge graph

Nodes + edges — people, topics, relationships.

Session logs

Verbatim transcript with speaker role.

Supabase Postgres + pgvector · 7 versioned, reversible Alembic migrations.

PRIVACY-FIRST — A FEATURE, NOT AN AFTERTHOUGHT

Your data stays yours.

Row-Level Security

- Enforced at the database
- A user can only ever touch their own graph & memories
- Sharing is opt-in, never default
- Personal growth tool — not org surveillance

PRODUCTION-GRADE OBSERVABILITY

We can see the system — from day one.

Sentry

errors

Prometheus

metrics

OpenTelemetry

traces

Grafana

dashboards

structlog

structured logs

HOW WE JUDGE SUCCESS

Honest numbers.

~300 ms	≥85%	low	~80%
advice latency	transcription target	hallucination rate	context-accurate answers

Across 30+ trial sessions (~350 transcript lines, 45+ graph relations) — and we're driving accuracy up.

ENGINEERING QUALITY

Discipline, not vibes.

- ✓ **mypy --strict** — fully typed Python
- ✓ **Ruff** — fast linting
- ✓ **pytest** — unit · integration · e2e
- ✓ **Locust** — load tested for concurrency
- ✓ **Docker multi-stage** — runtime + worker targets

HOW WE WORKED

Iterative R&D, clean ownership.

Research › Analysis › Design › Development › Testing

Ahmad

Client · architecture · data model · UI · releases

Attique

AI pipeline · memory · backend · workers · evaluation

CURRENT SCOPE

The honest edges of what's built today.

English-only	the language we support right now
Needs internet	the AI brain lives in the cloud
Thin client by design	the phone is mic + screen only

Design choices, not loose ends — a clear scope for what we demo today.

FUTURE PLANS · PLANNED, NOT YET BUILT

The runway.

1 **Multi-language**
Urdu first, then more

2 **Offline fallback**
local STT + small LLM (ONNX already in)

3 **Tone-aware coaching**
live sentiment → tone nudges

4 **Open-source release**
code + methods for the community

5 **Scale & deploy**
cloud GPU, app-store builds

6 **Fine-tuning**
domain-adapt ASR/LLM on sessions

Get better at talking —
one conversation at a time.

github.com/qdevaan/Bubbles-AI · **qdevaan.github.io/Bubbles-AI**

Thank you · Questions?